pharmaniaga®

Pharmaniaga Erythromycin Suspension 200 mg

DESCRIPTION:
Granules for Suspension.

Before reconstitution:A fine granular powder, practically free from visible impurities.

After reconstitution:Slightly viscous free flowing, homogeneous suspension, practically free from visible impurities.

COMPOSITION:
Each 5ml reconstituted suspension contains Erythromycin ethylsuccinate equivalent to Erythromycin 200 mg.

INDICATIONS:
Erythromycin is used in the treatment of infections due to susceptible organisms; it is often used as an alternative to penicillin in infections due to Gram-positive cocci, especially Streptococci, and *Listeria monocytogenes*. In penicillin sensitive patients it is given in place of penicillin in the prophylaxis of endocarditis and in the treatment of syphilis. For the treatment of otitis media caused by *Haemophilus influenzae* it may be given in combination with sulphonamide. Erythromycin may be given as an adjunct to antitoxin in the treatment of diphtheria or used alone to eradicate the carrier state. In the prevention of pertussis, it may be used in non- or partially immune patients. It is used in infections due to *Mycoplasma pneumoniae* and *Chlamydia trachomatis* in Legionnaires' disease, in enteritis due to *Campylobacter* spp., chancroid and similarly to tetracycline, in the treatment of severe acne vulgaris.

CONTRAINDICATIONS:
In patients with known hypersensitivity to erythromycin. Concurrent use with astemizole or terfenadine with erythromycin is contraindicated. The concurrent use may increase the risk of cardiotoxicity, such as *torsades de pointes* and ventricular tachycardia and death.

ACTIONS:
Erythromycin is a microlide antibacterial with a broad and essentially bacteriostatic action against many Gram-positive and to a lesser extent some Gram-negative bacteria, as well as other organisms including some *Mycoplasma* spp., Chlamydiaceae, *Rickettsia* spp., and spirochaetes. The actions of erythromycin are increased at moderately alkaline pH (up to about 8.5), particularly in Gram-negative species, probably because of the improved cellular penetration of the nonionised form of the drug.

PHARMACOKINETICS:
Absorption: Erythromycin base is unstable in gastric acid, and absorption is therefore variable and unreliable. Peak plasma concentrations generally occur between 1 and 4 hours after a dose and have been reported to range from about 0.3 to 1.0 micrograms/mL after 250mg of erythromycin base, and from 0.3 to 1.9 micrograms/mL after 500 mg. Peak concentration may be somewhat higher after repeated use 4 times daily.

Distribution: Erythromycin is widely distributed throughout body tissues and fluids, although it does not cross the blood-brain barrier well and concentrations in CSF are low. Relatively high concentrations are found in the liver and spleen. Erythromycin crosses the placenta: fetal plasma concentrations are variously stated to be 5 to 20% of those in the mother. It is distributed into breast milk.

Metabolism: Erythromycin is partly metabolised in the liver by the cytochrome P450 isoenzyme CYP3A4 via *N*-demethylation to inactive, unidentified metabolites.

Excretion: It is excreted in high concentrations in the bile and undergoes intestinal reabsorption. About 2 to 5% of an oral dose is excreted unchanged in the urine. The half-life of erythromycin is usually reported to be about 1.5 to 2.5 hours, although this may be slightly longer in patients with renal impairment

WARNINGS AND PRECAUTIONS:
Erythromycin should be given with caution in patients with impaired liver function. Erythromycin may potentiate the action of some drugs, including carbamazepine, cyclosporin, theophylline and warfarin probably by inhibition of their hepatic metabolism. Rare cases of serious cardiovascular adverse events including deaths, cardiac arrests, *torsades de pointes* and other ventricular arrhythmia have been observed, when used in patients taking concomitant terfenadine or astemizole.

There have been reports of infantile hypertrophic pyloric stenosis (IHPS) occurring in infants following erythromycin therapy. In 1 cohort of 157 newborns who were given erythromycin for pertussis prophylaxis, seven neonates (5%) developed symptoms of non-bilious vomiting or irritability with feeding and were subsequently diagnosed as having IHPS requiring surgical pyloromyotomy. Since erythromycin may be used in the treatment of conditions in infants which are associated with significant mortality or morbidity (such as pertussis or chlamydia), the benefit of erythromycin therapy needs to be weighed against the potential risk of developing IHPS. Parents and caregivers should be informed to contact their physician if vomiting and /or irritability with feeding occurs.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Steven-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) and acute generalised exanthematous pustulosis (AGEP)], Pharmaniaga Erythromycin Suspension 200 mg should be discontinued immediately and appropriate treatment should be urgently initiated.

DRUG INTERACTIONS:
Chronic preoperative or perioperative use of erythromycin may decrease the plasma clearance and prolong the duration of action of alfrentanil.

Erythromycin antagonises the effect of chloramphenicol and lincomycin: Concurrent use is not recommended.

Erythromycin has been reported to increase cyclosporin plasma concentrations and may increase nephrotoxicity.

Concurrent use of other hepatotoxic medications may increase the potential for hepatotoxicity.

Erythromycin used in patients who are receiving high doses of theophylline may be associated with an increase in serum theophylline levels and potential theophylline toxicity. In case of theophylline toxicity and/ or elevated serum theophylline levels, the dose of theophylline should be reduced while the patient is receiving concomitant erythromycin therapy.

Concomitant administration of erythromycin and digoxin has been reported to result in elevated digoxin serum levels. There have been reports on increased anticoagulant effects when erythromycin and oral anticoagulants were used concomitantly.

Concurrent use of erythromycin and ergotamine or dihydroergotamine has been associated in some patients with acute ergot toxicity characterised by severe peripheral vasospasm and dysaesthesia.

Erythromycin has been reported to decrease the clearance of triazolam and thus, may increase the pharmacological effect of triazolam.

The use of erythromycin in patients concurrently taking drugs metabolised by the cytochrome P450 system may be associated with elevations in serum erythromycin with carbamazepine, cyclosporin, hexabarbital and phenytoin. Serum concentrations of drugs metabolised by the cytochrome P450 system should be monitored closely in patients receiving erythromycin.

Patients receiving concomitant lovastatin and erythromycin should be carefully monitored; cases of rhabdomyolysis have been reported in seriously ill patients.

PREGNANCY AND LACTATION:
There are no adequate and well-controlled studies in pregnant women. However, observational studies in humans have reported cardiovascular malformations after exposure to medicinal products containing erythromycin during early pregnancy.

Erythromycin has been reported to cross the placental barrier in humans, but foetal plasma levels are generally low.

There have been reports that maternal macrolide antibiotics exposure within 7 weeks of delivery may be associated with a higher risk of infantile hypertrophic pyloric stenosis (IHPS).

Erythromycin can be excreted into breast-milk. Caution should be exercised when administering erythromycin to lactating mothers due reports of infantile hypertrophic pyloric stenosis in breast-fed infants.

ADVERSE REACTION:
Gastro-Intestinal disturbances are fairly common with erythromycin, especially with large doses; but serious side effects are rare. Supra-infection following oral administration of erythromycin is also rare, as is sensitisation to erythromycin. Skin reactions have been reported, but only isolated cases of anaphylaxis. Reversible deafness has occurred after high doses of erythromycin.

Postmarketing Experience:
Gastro-intestinal disorders: infantile hypertrophic pyloric stenosis.

Skin and Subcutaneous Tissue Disorders:
Frequency not known: severe cutaneous adverse reactions (SCARs) including Steven-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilis and systemic symptoms (DRESS) and acute generalised exanthematous pustulosis (AGEP).

DOSAGE AND ADMINISTRATION:
Adults:
For moderately severe infection, orally 1 g daily in 2 to 4 divided doses. For severe infections, 2 to 4 g may be given daily.

Children:
Orally 30 to 50 mg per kg of body weight daily; may be doubled in severe infections.

Children age 2 to 8 years: 1 g daily in divided doses.

Infants and children up to 2 years of age: 500 mg daily in divided doses.
Route of Administration: Oral

USER INSTRUCTIONS:
1. Reconstitution:
   Shake the bottle to loosen the powder. Add freshly boiled and cooled water and shake well. Make up to volume. Use reconstituted suspension within 7 days.
2. Take suspension at regular intervals preferably after meals.
3. Complete the prescribed course.
4. Each time before use, shake the container well.

OVERDOSAGE:
Symptoms: Extended effect of the side-effects.

Treatment: Stop therapy, and give supportive and symptomatic treatment.

STORAGE CONDITIONS:
Store below 30°C in a cool dry place; Protect from light and moisture. Keep in refrigerator after reconstitution (2°C to 8°C).

SHELF LIFE:
Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION:
Bottles of 60 mL and 100 mL.

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Lt-203.03 Product Registration Holder and Manufacturer

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